

# Run report

2026-10-04T022859Z

## Configuration

| Parameter           | Value                                                                                     | Parameter     | Value                  |
|---------------------|-------------------------------------------------------------------------------------------|---------------|------------------------|
| batch_size          | 1,024                                                                                     | n_samples     | 500,000                |
| dim                 | 1                                                                                         | dataset       | gaussian               |
| seed                | 1,175,258,683                                                                             | data_seed     | 42                     |
| hidden_units        | 64                                                                                        | n_layers      | 2                      |
| n_epochs            | 100                                                                                       | n_disc_steps  | 5                      |
| lr_g                | $3.000 \times 10^{-5}$                                                                    | lr_d          | $1.000 \times 10^{-4}$ |
| lambda_dispersion   | 0.01500                                                                                   | mmd_subsample | 16,384                 |
| mmd_test            | $5.379 \times 10^{-3}$                                                                    | best_epoch    | 98                     |
| compile_cache_warm  | True                                                                                      |               |                        |
| gaussian_params     | dim=1, mu_gen=[0.5], mu_true=[0], cov_gen=[[0.81]], cov_true=[[1]], cov_detector=[[0.25]] |               |                        |
| mmd_sigmas_detector | $0.7470 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$                         |               |                        |
| mmd_sigmas_particle | $0.6688 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$                         |               |                        |

## Timing

| Phase    | Seconds | Share (%) | Detail                    |
|----------|---------|-----------|---------------------------|
| data     | 0.051   | 1.0       | gaussian cache hit – load |
| load     | 1.474   | 28.6      | load                      |
| plots    | 3.627   | 70.3      | load                      |
| evaluate | 0.005   | 0.1       | load                      |
| total    | 5.156   | 100.0     |                           |

## Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

### Wasserstein Distance [ $\times 10^3$ ]

| Observable | Sim.  | IBU   | impr. (%) | OmniFold     | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|--------------|-------------|-------|-----------|
| dim_0      | 507.8 | 18.41 | 96.4      | <b>13.13</b> | <b>97.4</b> | 97.77 | 80.7      |

### JS divergence [ $\times 10^3$ ]

| Observable | Sim.  | IBU           | impr. (%)   | OmniFold | impr. (%) | RAN   | impr. (%) |
|------------|-------|---------------|-------------|----------|-----------|-------|-----------|
| dim_0      | 28.91 | <b>0.5955</b> | <b>97.9</b> | 0.7157   | 97.5      | 1.792 | 93.8      |

### VLC divergence [ $\times 10^3$ ]

| Observable | Sim.  | IBU          | impr. (%)   | OmniFold | impr. (%) | RAN   | impr. (%) |
|------------|-------|--------------|-------------|----------|-----------|-------|-----------|
| dim_0      | 111.5 | <b>2.165</b> | <b>98.1</b> | 2.624    | 97.6      | 6.878 | 93.8      |

### Sliced Wasserstein, joint [ $\times 10^3$ ]

| Observable | Sim.  | IBU   | impr. (%) | OmniFold     | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|--------------|-------------|-------|-----------|
| <i>All</i> | 455.0 | 16.49 | 96.4      | <b>11.76</b> | <b>97.4</b> | 87.60 | 80.7      |

Mean 1D distance over random directions, in Data standard deviations. IBU: product of its per-observable weights.

## Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

### Wasserstein [ $\times 10^3$ ]

| Observable | Gen   | IBU   | impr. (%) | OmniFold     | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|--------------|-------------|-------|-----------|
| dim_0      | 508.5 | 42.23 | 91.7      | <b>12.35</b> | <b>97.6</b> | 97.44 | 80.8      |

### JS divergence [ $\times 10^3$ ]

| Observable | Gen   | IBU   | impr. (%) | OmniFold      | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|---------------|-------------|-------|-----------|
| dim_0      | 37.03 | 7.828 | 78.9      | <b>0.7511</b> | <b>98.0</b> | 2.432 | 93.4      |

### VLC divergence [ $\times 10^3$ ]

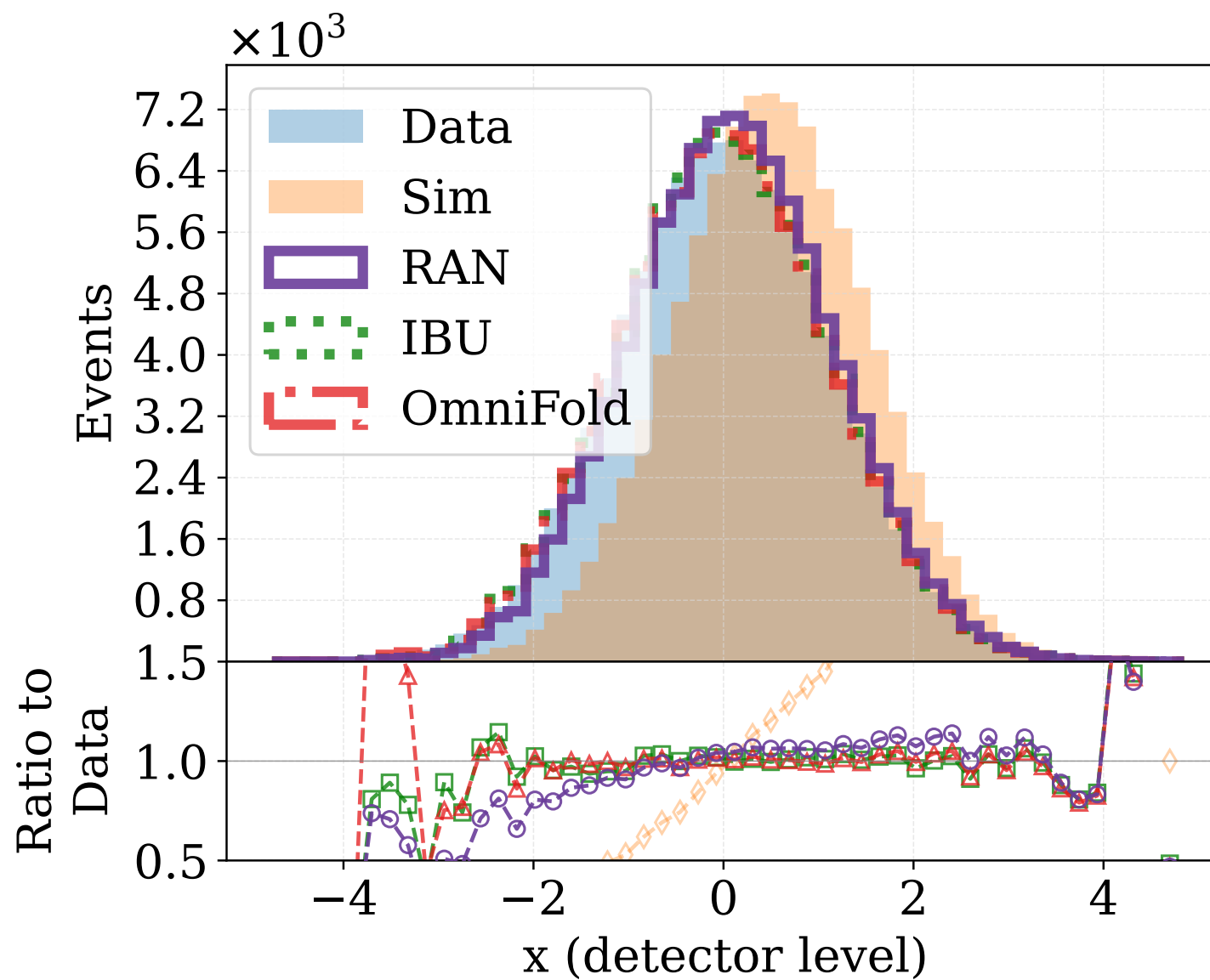
| Observable | Gen   | IBU   | impr. (%) | OmniFold     | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|--------------|-------------|-------|-----------|
| dim_0      | 141.3 | 30.88 | 78.2      | <b>2.778</b> | <b>98.0</b> | 9.347 | 93.4      |

### Sliced Wasserstein, joint [ $\times 10^3$ ]

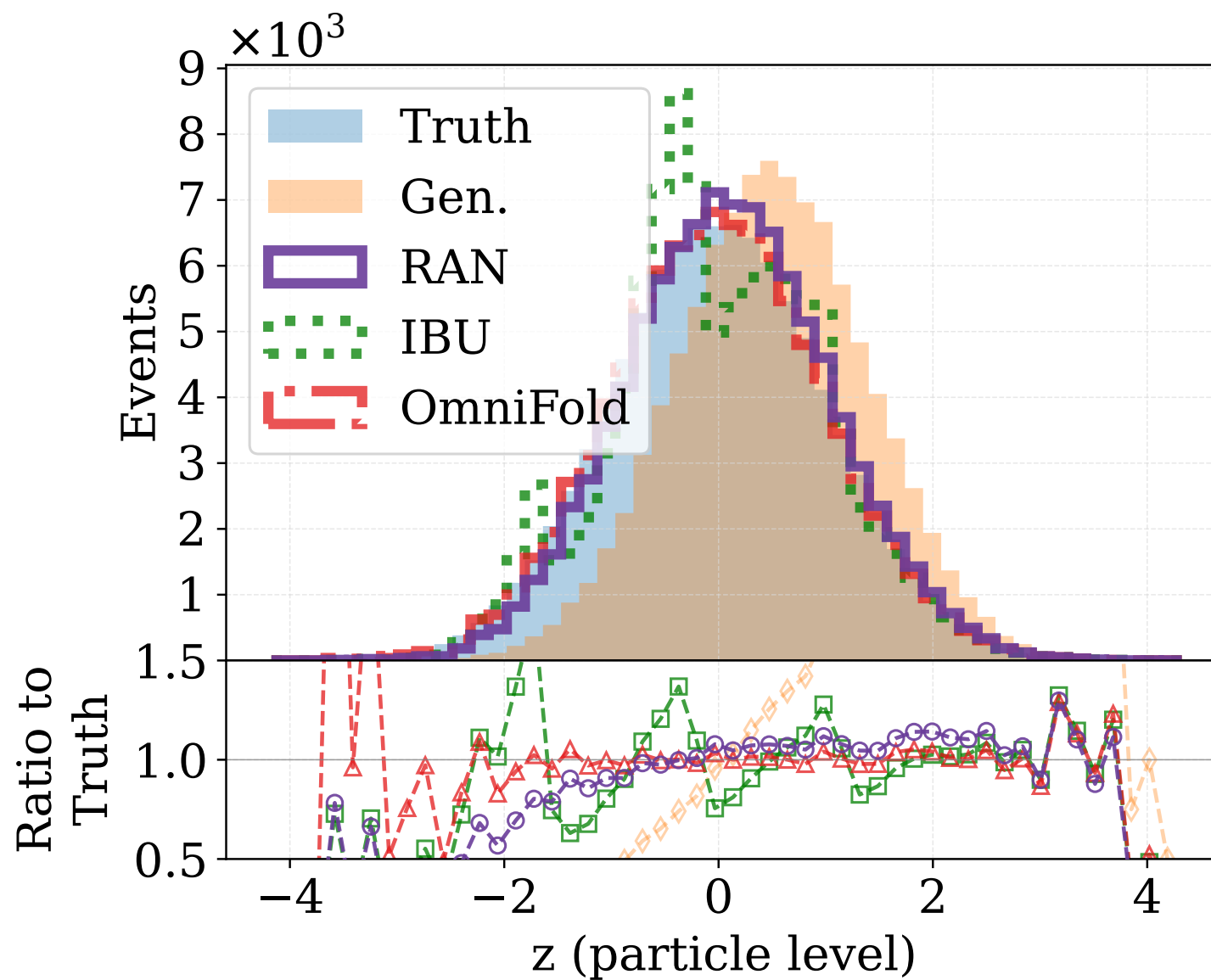
| Observable | Gen   | IBU   | impr. (%) | OmniFold     | impr. (%)   | RAN   | impr. (%) |
|------------|-------|-------|-----------|--------------|-------------|-------|-----------|
| <i>All</i> | 508.4 | 42.23 | 91.7      | <b>12.34</b> | <b>97.6</b> | 97.44 | 80.8      |

Mean 1D distance over random directions, in Truth standard deviations. IBU: product of its per-observable weights.

# Detector Level

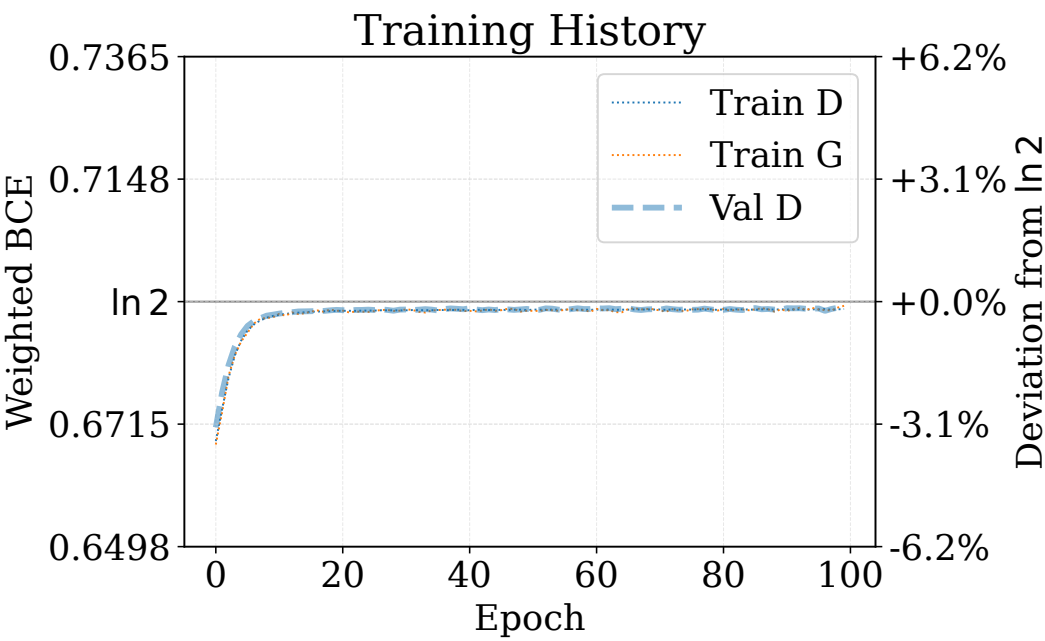


# Particle Level



Training diagnostics

Training losses



Selection criteria

